

Technology Stack

Date	30 June 2026
Team ID	
Project Name	Smart Lender Loan Eligibility Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	<i>HTML, CSS, JavaScript, Bootstrap</i>	Used for building a responsive and user-friendly interface where applicants can enter loan details and view predictions. Bootstrap ensures quick styling and mobile compatibility.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python (Flask Framework)	Flask provides lightweight, fast, and flexible server-side development. It integrates seamlessly with machine learning models and supports RESTful APIs for real-time predictions.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	MySQL	<i>Reliable relational database for storing applicant data, prediction logs, and user credentials. Ensures structured data management and easy integration with Flask.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	IBM Cloud	Chosen for its scalability, security, and ease of deployment. Supports containerized applications and integrates well with Python-based environments.

5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	GitHub + Git	Enables collaborative development, version tracking, and continuous integration. Ensures smooth updates and rollback capabilities.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Scikit-learn, XGBoost, Pandas, NumPy	Scikit-learn, XGBoost, Pandas, NumPy